HUSH: Holistic Panoramic 3D Scene Understanding using Spherical Harmonics

Jongsung Lee[1], Harin Park[1], Byeong-Uk Lee[2], Kyungdon Joo[1*]
[1]Artificial Intelligence Graduate School, UNIST, [2]KRAFTON
{syniez, harinp33, kyungdon}@unist.ac.kr
byeonguk.lee@krafton.com

Abstract

Motivated by the efficiency of spherical harmonics (SH) in representing various physical phenomena, we propose a Holistic panoramic 3D scene Understanding framework using Spherical Harmonics, dubbed as `HUSH`*. Our approach focuses on a unified framework adaptable to various 3D scene understanding tasks via SH bases. To achieve this, we first estimate SH coefficients, allowing for the adaptive configuration of the SH bases specific to each scene.* `HUSH` *then employs a hierarchical attention module that uses SH bases as queries to generate comprehensive scene features by integrating these scene-adaptive SH bases with image features. Additionally, we introduce an SH basis index module that adaptively emphasizes relevant SH bases to produce task-relevant features, enhancing the versatility of* `HUSH` *across different scene understanding tasks. Finally, by combining the scene features with task-relevant features in the task-specific heads, we perform various scene understanding tasks, including depth, surface normal and room layout estimation. Experiments demonstrate that* `HUSH` *achieves state-of-the-art performance on depth estimation benchmarks, highlighting the robustness and scalability of using SH in panoramic 3D scene understanding. For more information, you can visit our project page* https://vision3d-lab.github.io/hush/.

1. Introduction

The 360° camera is becoming more popular to be used in various fields of computer vision and robotics, as it can capture a 360° view of an environment within a single frame due to its large field of view (FoV). The capability to provide a complete spatial context makes it highly advantageous in understanding holistic 3D scenes. As a result, there is a dramatic increase of interest in 3D understanding tasks like depth estimation or surface normal estimation using 360° images as input.

The most popular way of interpreting 360° images is equirectangular projection (ERP). It maps 360° view to a 2D plane, and is commonly known as panorama images. ERP image is advantageous in a way that it holds all the information in a single image, but it contains visual distortion, especially near the pole regions. This makes an ERP image difficult to be utilized in conventional computer vision algorithms since most of the methods are trained on perspective images. Therefore, many of the previous approaches try to simplify the process by converting ERP images into multiple perspective patches, as in cubemap projections (CP) [12, 34] or tangent projections (TP) [3, 19, 26]. Another line of works focuses on utilizing ERP image itself, by either assuming strong geometric constraints or developing ERP-specific convolution modules [28, 38, 39].

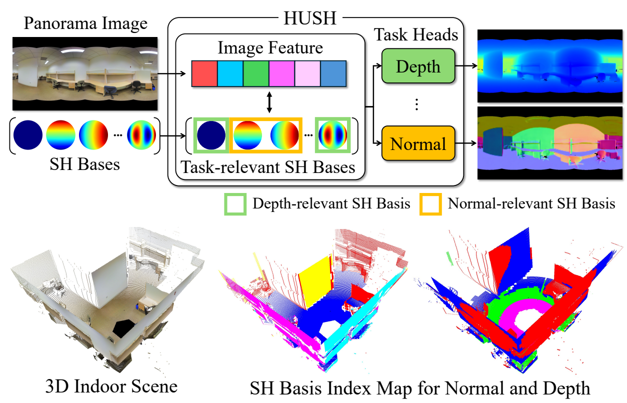

Figure 1. **Illustration of `HUSH`**. Our method exploits SH bases to perform various panoramic 3D scene understanding tasks, such as estimating depth and surface normals. In particular, `HUSH` focuses more on the SH bases relevant to each task. The bottom illustrates an example of the SH basis index map for each task, where we can observe colorized task-relevant SH index maps. For visualization purposes only, we use the ground truth depth map.

While patch-based approaches show promising results in panorama-based 3D scene understanding tasks, they commonly suffer from computational inefficiency due to constant conversion between an ERP image and multiple perspective images. In this work, we focus on the nature of ERP. Panoramic images are inherently represented in a

*Corresponding author.

spherical coordinate system. Estimating pixel-wise information on ERP images is equivalent to the estimation of the information on a spherical surface. This can be efficiently approximated by spherical harmonics (SH) [25], a mathematical framework that decomposes functions on the sphere into a series of basis functions. Our insight is that by utilizing SH, we can implicitly approximate various task-relevant features across multiple basis functions, facilitating a more comprehensive understanding of 3D spherical structures.

Based on this insight, we newly introduce a `H`olistic panoramic 3D scene `U`nderstanding framework based on `S`pherical `H`armonics, `HUSH`. As shown in Fig. 1, it utilizes SH basis functions as implicit structural guidance for comprehensive scene understanding. To successfully aggregate the image features with the SH bases, we adopt global context-aware feature extraction architectures with hierarchical cross-attention module and SH basis index module. Specifically, for the hierarchical cross-attention, we use global image features and SH bases as key-values and queries, respectively. This integration of SH, transformer-based architecture, and SH-based attention mechanism allows a robust 3D interpretation of 360° data.

Our proposed pipeline aims to enable comprehensive 3D scene understanding rather than only focusing on a single task. Thanks to the utilization of SH, our method provides a scalable way to approximate features necessary for different 3D tasks, improving both performance and generalizability. To demonstrate the effectiveness of our approach, we conduct a series of experiments that examine the interpretability of our network and resulting SH-based 3D understanding. To summarize, our main contributions are:

- We propose a comprehensive 3D scene understanding framework based on SH that supports diverse 3D tasks on panoramic data, including depth estimation, surface normal estimation, indoor layout prediction, and more.
- We introduce an SH-based hierarchical attention mechanism that approximates task-specific features across multiple SH basis functions and successfully integrates SH into the panorama understanding pipeline.
- We conduct extensive experiments to validate our insight, design, and interpretability and show the robustness and scalability of utilizing SH for panoramic scene understanding.

2. Related Work

Monocular depth estimation. Monocular depth estimation has shown remarkable performance, driven by breakthroughs in CNN and transformers. In recent trends, several methods utilize a combination of regression and classification, making training easier and ensuring stable performance. These approaches estimate depth by a linear combination of depth planes and the probabilities that each pixel belongs to a specific plane, thereby effectively considering the structural properties of the scene [5, 6, 18].

While these depth plane-based methods focus on explicitly dividing the depth range, some recent methods implicitly learn the internal representations of scenes using learnable queries. For example, iDisc [24] introduces an internal discretization module that partitions the scene into high-level patterns using learnable queries. Similarly, Plane2depth [20] proposes a hierarchical adaptive framework that utilizes learnable plane queries to model plane information effectively. These implicit approaches can be expanded to various related tasks beyond depth estimation, as demonstrated by iDisc.

Panorama depth estimation. Estimating depth from a single panorama image is quite challenging due to the severe distortions from 360° cameras. To mitigate such distortions, several methods [28, 38, 39] propose panorama-specific transformer architectures for large and flexible receptive fields. While these approaches technically handle panoramic distortions, the other methods use projection strategies such as CP that can consider both local details and holistic information [12, 34]. UniFuse [12] utilizes CP for 360° depth estimation and fuses cubemap depth with ERP depth at the end of the network. BiFuse [34] addresses distortion and boundary discontinuity by integrating ERP and CP, utilizing learnable masks and spherical padding.

A recent trend in 360° depth estimation is using more dense representations, such as TP and icosahedron projection (ICOSAP). 360MonoDepth [26] focuses on the high-resolution depth estimation using tangent patches, aiming to mitigate discrepancy on the overlapped regions. Despite this effort, the discrepancy problem remains, and OmniFusion [19] proposes a geometric embedding network to solve this issue. Panoformer [28] also uses TP with panoramic computations (*e.g.*, relative positional embedding, panoramic self-attention) to effectively model geometric structures. Although these TP-based 360° depth estimation methods are well-performed, geometric projections (ERP $\leftrightarrow$ TP, CP) consume high computational cost. HRDFuse [3] argues this problem and proposes a spatial feature alignment module that efficiently aggregates TP features into the ERP domain. Using two features from ERP and ICOSAP domain, Elite360D [1] successfully learns global-local receptive fields for large FoV images.

Panorama indoor scene understanding. Panoramic images provide significant advantages in indoor scene understanding by capturing a complete 360° view with rich structural and contextual information. This unique capability has led to their use not only in depth estimation but also in various scene understanding tasks, such as surface normal and room layout estimation [2, 9, 10, 32, 38]. For example, HoHoNet [32] introduces latent horizontal features to capture scene-level structures, which facilitate depth and layout es-

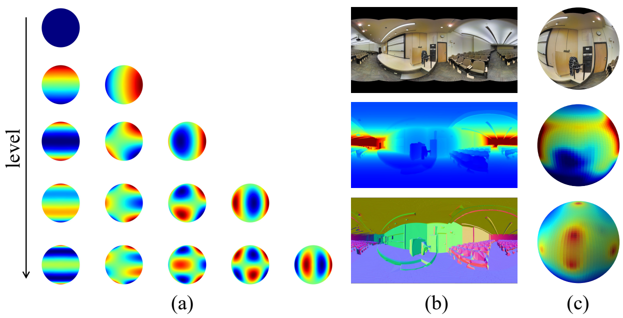

Figure 2. **Visualization of SH bases and data distributions on spheres**. (a) A set of SH basis functions. (b) Example of RGB scene with ground truth (depth and normal) in the ERP domain. (c) RGB mapping and depth/normal distributions on the unit sphere.

timation, among other tasks. PanelNet [38] also performs various scene understanding tasks by dividing ERP images into panels, effectively capturing the global and local structures. These studies show that the features capturing the structure of panoramic images can be effectively utilized for scene understanding tasks. However, since these methods train each task independently, the exchange of information across complementary tasks can be limited. In response to this, recent studies have suggested approaches that enable the joint learning of diverse scene understanding tasks [2, 9]. PanoContext-Former [9] designs a transformer-based context module to predict relationships among scene components while performing multiple tasks, including layout estimation. In addition, Elite360M [2] proposes a cross-task collaboration module to facilitate information exchange among tasks, resulting in enhanced performance.

3. Method

3.1. Motivation

Spherical harmonics (SH) are a set of orthogonal functions defined on the surface of a sphere [25]. They have been a popular choice of low-dimensional representation for spherical functions, such as in approximating Lambertian surfaces [11, 36] and view-dependent radiance in novel view synthesis approaches [16, 35, 37]. Inspired by this fact, we focus on the potential of SH in panoramic 3D scene understanding that covers 360° FoV of a given indoor scene.

Equirectangular projections (ERP) are a way of representing 360° capture by mapping pitch to vertical lines and yaw to horizontal lines. It enables one-to-one mapping between panorama images and 3D unit spheres. Thus, if we consider ERP of depth information (*i.e.*, panorama depth image) as a spherical function, then we may utilize SH to approximate the information. The same insight can apply to any kind of information, such as surface normal. On top of that, the orthogonality of SH basis functions naturally aligns with 3D geometry of indoor scenes. Utilizing a pre-defined orthogonal basis for 3D scene understanding is similar to conventional geometric assumptions made in 3D understanding, such as Manhattan [8, 14] or Atlanta [15, 27] world assumptions. Figure 2 illustrates the depth and normal distributions on a sphere. Here, we can intuitively observe the connection between the SH bases and 3D understanding tasks for panorama images.

This insight of relating panoramic 3D geometry to SH is the main motivation for our `H`olistic panoramic 3D scene `U`nderstanding pipeline using `S`pherical `H`armonics, `HUSH`. In the following sections, we describe how we design this novel insight as a deep learning architecture and an attention mechanism and how we validate our design choices.

3.2. Overall Pipeline

`HUSH` takes a single panorama image as an input and predicts various 3D representations with respect to the training configuration. As illustrated in Fig. 3, our framework first extracts multi-scale image features using pre-trained swin transformer [21] encoder with multi-scale deformable attention (MSDA) [41] blocks. The extracted features and intermediate embeddings are then fed into our novel spherical harmonics attention mechanism (Sec. 3.3), which is designed to effectively aggregate scene and task-wise SH bases to the context-aware image features, in order to obtain comprehensive scene features and task-relevant features. Our SH attention mechanism consists of scene-wise SH coefficient extractor, SH-based hierarchical attention module and task-wise SH basis index module. Finally, our task-specific prediction heads take SH integrated features as input and estimate 3D information such as depth map, surface normal, and room layout (Sec. 3.4).

3.3. Spherical Harmonics Attention Mechanism

SH bases are designed to approximate functions defined on spherical surfaces. The most common and traditional way is to calculate SH coefficients, which allows the desired final value or function to be represented as a weighted summation of pre-defined SH basis functions. Beyond this conventional use, we propose that it may be more beneficial to treat SH bases as implicit structural prior and leverage them as query values for attention mechanisms, rather than solely using them as approximation tools.

To do so, we introduce a spherical harmonics attention mechanism. Our attention framework first scales pre-defined SH bases via a scene-wise SH coefficient extractor so that the modified bases are more aligned with the input scene. Then, it extracts comprehensive scene features using an SH-based hierarchical attention module, utilizing modified SH basis functions as queries. At the same time, our task-wise SH basis index module obtains task-relevant features by amplifying the role of the most dominant SH basis depending on the scene, region, and task. By virtue of our

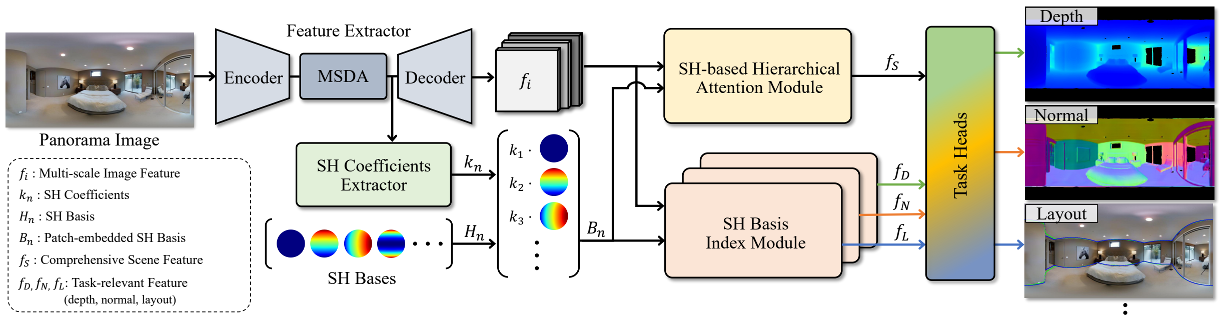

Figure 3. **Overview of the proposed `HUSH` framework**. `HUSH` first extracts multi-scale image features f_i and scene-wise SH bases via feature extractor and SH coefficients network. These image features and the scene-wise SH bases are then fed into the SH-based hierarchical attention module and the SH basis index module to estimate comprehensive scene feature f_S and task-relevant features (f_D, f_N, f_L). Finally, various scene-understanding tasks are performed as these features pass through task-specific heads.

SH attention mechanism, we effectively fuse information from multi-scale image features and SH basis functions.

Scene-wise SH coefficients extractor. Each input panorama image holds a different scene structure. In order to utilize SH as more effective attention queries, we first modify pre-defined SH bases so that the new bases are more adaptively set according to each scene.

Our SH coefficient extractor predicts SH coefficients using image features from the MSDA module. The obtained SH coefficients are multiplied to the original SH basis functions. The modified SH basis functions are scene-wise scaled versions of the original SH bases while maintaining orthogonality, making them more scene-adaptive structural priors for the following hierarchical attention module.

SH-based hierarchical attention module. In the recent work, Liu *et al*. [20] first propose a hierarchical adaptive framework, Plane2Depth, for monocular depth estimation using plane queries. It successfully aggregates multi-scale features with its learnable plane queries for estimating the plane coefficients from a given scene. The proposed hierarchical attention-based mechanism emphasizes the guidance of global context features and ensures the consistency of multi-scale features in query aggregation.

We adopt a similar hierarchical attention mechanism into our framework, expecting that our framework can effectively aggregate the multi-scale image features f_i $(i \in \{1, 2, 3\})$ with our SH bases H_n $(n \in \{1, 2, ..., 55\})$ while keeping the global context. Our proposed SH-based hierarchical attention module mainly consists of several cross-attention blocks. However, unlike the previous approach, we utilize target task-specific queries (*i.e*., modified SH basis functions) rather than randomly initialized learnable queries. This would provide better panoramic scene understanding guidance based on our motivation mentioned in Sec. 3.1. Comparison between using learnable queries and SH queries is described in Sec. 4.3.

We use patch-embedded modified SH bases B_n as queries and feed them to the first cross-attention block

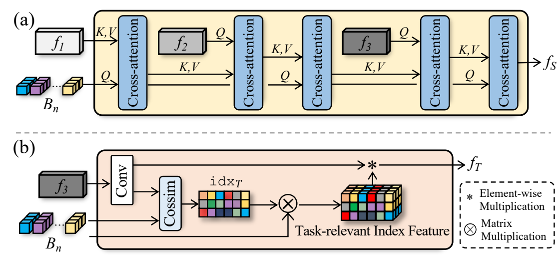

Figure 4. **Details of the proposed modules**. (a) SH-based hierarchical attention module. (b) SH basis index module.

alongside multi-scale image features f_i. By feed-forwarding through multiple cross-attention blocks, the attention features and SH bases alternate roles as queries and key-values. As a result, we can obtain the comprehensive scene feature f_S, which inherently includes both scene context and spherical geometries. The detailed process of the hierarchical attention mechanism is described in Fig. 4.

Task-wise SH basis index module. The core motivation behind our proposed pipeline is that SH basis functions would provide comprehensive geometric guidance of the scene. However, depending on the task and the region, there may be a difference in which basis plays a more dominant role. Therefore, our goal is to provide task-specific pixel-wise guidance on which basis to focus more, while keeping the total number of SH bases sufficient to maintain comprehensive 3D scene representation.

To this end, we design an SH basis index module, which is similar to the spatial feature alignment (SFA) module in [3]. In the SH basis index module, it first generates task-relevant feature f_T by simply passing the last feature of image features f_i to the single convolution layer. Here, f_T collectively represents task-relevant features for depth (f_D), normal (f_N), and layout (f_L). Then, it computes task index map $\mathtt{idx}_T$, based on cosine similarity between the patch-embedded SH bases B_n and the task-relevant feature f_T as,

$$\mathtt{idx}_T(u, v) = \arg\max_n \left(\frac{f_T(u, v) \cdot B_n}{\|f_T(u, v)\| \, \|B_n\|}\right), \quad (1)$$

where (u, v) is the pixel coordinate and n is the SH basis index. This index map acts as an attention map and composes task-relevant index feature that is multiplied to the task-relevant feature f_T, so that the final feature is more concentrated on the most dominant SH basis. Please refer to Fig. 4 for more details.

Our SH basis index module makes each pixel focus more on a certain SH basis, task-wise. Our intuition is that this will make each SH basis to be more dominant than the others based on the task, the scene and the region. This will separate them task-wise, and enable robust 3D scene understanding by also relieving ambiguity when calculating attention among a large number of SH bases. Visualization of our basis index map and validation of our intuition are further analyzed in Sec. 4.3.

3.4. Task-wise 3D Scene Understanding

Now that we have obtained SH-guided comprehensive scene features and task-relevant features, we utilize them for robust 3D scene understanding. Our `HUSH` framework supports multi-task 3D understanding with the shared image feature extraction pipeline and the SH-based attention module. While our pipeline holds extendability to various types of 3D understanding tasks, our baseline model is trained to solve the two most common 3D problems: depth estimation and surface normal estimation.

The architecture design of each task-wise prediction head is free of choice. Yet, our depth head only consists of a single cross-attention block and four linear layers, thanks to our SH-based attention mechanism, which extracts comprehensive features. The depth head utilizes the comprehensive feature f_S and depth-relevant feature f_D obtained using the task-specific SH basis index module. The surface normal head shares the same design as our depth head, except its last layer is modified to output three channels of information. Note that a separate SH basis index module is defined for the surface normal prediction head since bases may be weighted differently for different tasks.

3.5. Loss Function

Depth loss. For depth estimation, we follow the previous methods [1, 3, 19] and adopt reverse Huber (BerHu) loss [17] function:

$$\mathcal{B}(e) = \begin{cases} |e| & |e| \leq c \\ \frac{e^2+c^2}{2c} & |e| > c \end{cases}, \tag{2}$$

where e is the error term and the threshold c decides where the L1 and L2 loss changes. Then, we can compute the depth loss between ground truth depth D_m and the depth prediction $\tilde{D}_m$ on M valid pixels:

$$\mathcal{L}_{depth} = \frac{1}{M}\sum_{m=1}^{M}\mathcal{B}(D_m - \tilde{D}_m). \tag{3}$$

Gradient loss. Additionally, we also use spherical gradient loss in [30], to prevent smoothing effects from discrete SH queries and sharpen the object boundaries:

$$\begin{aligned} \mathcal{L}_{grad} = \frac{1}{M}\sum_{m=1}^{M}(&|G_x(D_m) - G_x(\tilde{D}_m)| \\ &+ |G_y(D_m) - G_y(\tilde{D}_m)|), \end{aligned} \tag{4}$$

where G_x and G_y denote gradients along x and y directions for ground truth depth and the predicted depth.

Normal loss. For surface normal estimation, we use cosine similarity loss function to ground truth surface normal and predicted surface normal, which is represented as a unit vector:

$$\mathcal{L}_{norm} = \frac{1}{M}\sum_{m=1}^{M}(1 - N_m \cdot \tilde{N}_m), \tag{5}$$

where N and $\tilde{N}$ denotes ground truth surface normal and predicted surface normal.

Total loss. Finally, the total loss for our problem $\mathcal{L}_{total}$ is:

$$\mathcal{L}_{total} = \mathcal{L}_{depth} + \mathcal{L}_{norm} + \lambda\mathcal{L}_{grad}, \tag{6}$$

where λ is a weight factor set to 0.5 for our experiment.

4. Experiment

4.1. Datasets, Metrics and Implementation Details

Dataset. For evaluation, we use three panorama benchmark datasets: Matterport3D [7], Stanford2D3D [4], and Structured3D [40]. Matterport3D and Stanford2D3D are real-world panorama datasets that contain 10,790 and 1,413 panorama RGBD images from 90 scenes and 6 large-scale indoor areas, respectively. Structured3D provides large-scale indoor synthetic 2D rendering with rich 3D annotations. To ensure consistency of the training procedure, we resize all the dataset input RGB panoramas to a resolution of 512 $\times$ 1024. We utilize computed normals using the ground truth depths on the real-world datasets and given ground truth surface normals on the synthetic dataset.

Evaluation metrics. To evaluate the depth estimation performance, we use the standard depth estimation metrics used in the previous work [1]: absolute relative error (Abs Rel), squared relative error (Sq Rel), root mean square error (RMSE), and three threshold-based accuracy $\delta_i = 1.25^i$ for $i \in \{1, 2, 3\}$.

Implementation details. We implement `HUSH` using Pytorch and train it on a single NVIDIA 4090 GPU with a batch size of 2. We use AdamW [23] optimizer with weight decay 10^{-2}, and 1-cycle policy [31] for the learning rate with max lr = 3.5×10^{-4}. For the first 30% of iterations, the linear learning rate warm-up technique is utilized. The cosine annealing [22] learning rate policy is applied for the learning rate decay.

Datasets	Method	Abs Rel↓	Sq Rel↓	RMSE↓	δ_1↑	δ_2↑	δ_3↑
Stanford2D3D [4]	UniFuse [12]	0.1124	0.0709	0.3555	0.8706	0.9704	0.9899
	PanoFormer [28]	0.1122	0.0786	0.3945	0.8874	0.9584	0.9859
	HRDFuse* [3]	0.1052	0.0739	0.3725	0.8962	0.9701	0.9883
	EGformer [39]	0.1528	0.1408	0.4974	0.8185	0.9338	0.9736
	Elite360D [1]	0.1182	0.0728	0.3756	0.8872	0.9684	0.9892
	Ours (D)	**0.0783**	**0.0548**	**0.3418**	**0.9375**	**0.9843**	**0.9929**
	Elite360M* [2]	0.1014	0.0630	0.3605	0.8979	0.9688	0.9907
	Ours (D+N)	**0.0782**	**0.0547**	**0.3332**	**0.9384**	**0.9849**	**0.9929**
Matterport3D [7]	UniFuse [12]	0.1144	0.0936	0.4835	0.8785	0.9659	0.9873
	PanoFormer [28]	0.1051	0.0966	0.4929	0.8908	0.9623	0.9831
	HRDFuse [3]	0.1172	0.0971	0.5025	0.8674	0.9617	0.9849
	HRDFuse* [3]	0.1082	0.1032	0.4728	0.9042	0.9591	0.9803
	EGformer [39]	0.1473	0.1517	0.6025	0.8158	0.9390	0.9735
	Elite360D [1]	0.1115	0.0914	0.4875	0.8815	0.9646	**0.9874**
	Ours (D)	**0.0845**	**0.0821**	**0.4167**	**0.9291**	**0.9706**	0.9858
	Elite360M [2]	0.1178	-	-	0.8672	0.9627	-
	Elite360M* [2]	0.0946	0.0835	0.4642	0.9095	0.9656	0.9850
	Ours (D+N)	**0.0838**	**0.0815**	**0.4164**	**0.9287**	**0.9698**	**0.9851**
Structured3D [40]	UniFuse [12]	0.1506	0.2319	0.5016	0.8542	0.9399	0.9676
	PanoFormer [28]	0.2549	0.4949	0.7937	0.7470	0.8915	0.9397
	HRDFuse* [3]	0.1639	0.2726	0.5418	0.8458	0.9353	0.9622
	EGformer [39]	0.2205	0.4509	0.6841	0.7979	0.9071	0.9455
	Elite360D [1]	0.1480	0.2215	0.4961	0.8741	0.9434	0.9666
	Ours (D)	**0.0503**	**0.0459**	**0.2852**	**0.9728**	**0.9910**	**0.9955**
	Elite360M [2]	0.1430	-	-	0.8671	0.9511	-
	Elite360M* [2]	0.1505	0.2111	0.5412	0.8486	0.9442	0.9735
	Ours (D+N)	**0.0459**	**0.0396**	**0.2577**	**0.9776**	**0.9922**	**0.9961**

Table 1. **Quantitative comparison of depth estimation with the SOTA methods**. **Bold** indicates the best performance. * denotes reproduced result. Ours (D) represents depth-only training, and Ours (D+N) means multi-task learning of depth and surface normal.

4.2. Comparison

Considering the difference in the experimental setup, we follow the recent state-of-the-art method [1]. Following [1], we train HUSH for 100 epochs on the real-world datasets, Matterport3D and Stanford2D3D, and 30 epochs on the synthetic dataset, Structured3D.

Since we can train our model with various modalities, we compare HUSH with recent panorama depth estimation methods: UniFuse [12], PanoFormer [28], HRDFuse [3], EGFormer [39], Elite360D [1] and the multi-task method: Elite360M [2]. For HRDFuse, we also report reproduced results based on the public code, and denote it as HRDFuse*. For Elite360M, which was initially designed for the multi-task of depth, surface normal and semantic segmentation, we report additional variant that is trained only on depth and surface normal estimation, to make it a fair comparison with our model, and denote it as Elite360M*.

Depth estimation. As shown in Table 1, HUSH outperforms the other panorama depth estimation methods on all three benchmark datasets by a large margin. In particular, RMSE and AbsRel metrics are significantly improved; 14.52% and 24.1% compared to Elite360D on the Matterport3D dataset. Once we can extend HUSH to the multi-task framework, we train HUSH with depth and surface normal guidance and compare it with the recent panorama multi-task framework, Elite360M*. Remarkably, HUSH also outperforms Elite360M* with 10.29% and 11.43% improvements in RMSE and AbsRel on the Matterport3D dataset.

Figure 5 shows the qualitative comparison of HUSH with two state-of-the-art methods: HRDFuse* for depth estimation and Elite360M* for panorama multi-task framework. We can observe that comparison methods show blurry depth boundaries, especially on the texture regions, because they adopt projection-based methods (*e.g.*, TP and ICOSAP). This could lead to disadvantages when they merge the depth results, since seamless fusions between depths from the patch and the ERP could lead to over-smoothing. On the other hand, thanks to the SH-based unified 3D scene understanding framework, HUSH generally shows accurate depth predictions, even with sharp object boundaries.

Surface normal estimation. Figure 6 shows the qualitative results of surface normal estimation by HUSH and Elite360M*. Our method shows more detailed surface normal results with consistency on the plane region, while Elite360M* suffers from over-smoothing and inconsistency. Additional qualitative results and quantitative evaluation for surface normal are available in the supplementary material.

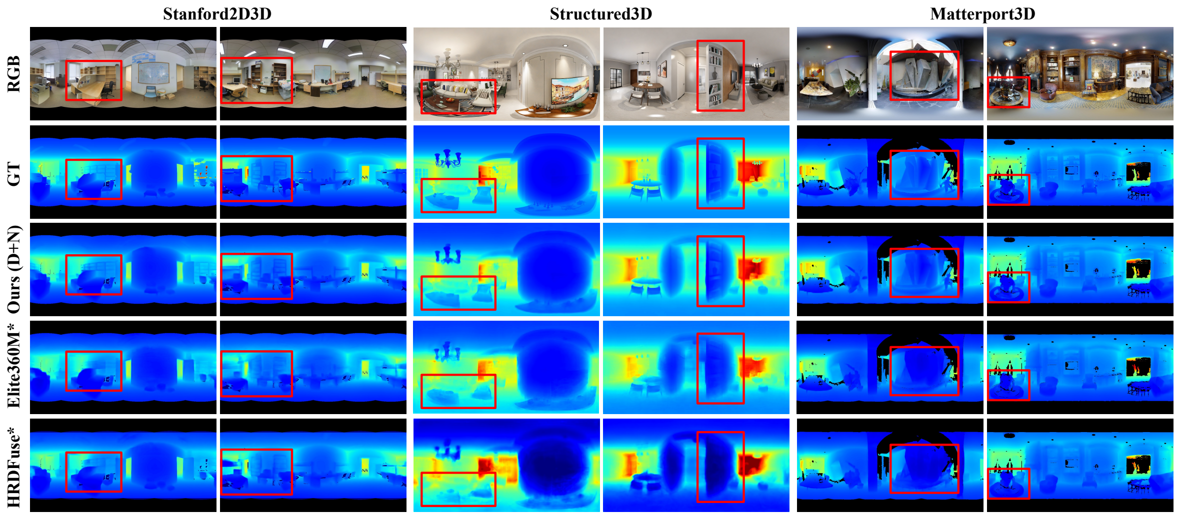

Figure 5. **Qualitative comparison of depth estimation on Stanford2D3D, Structured3D, and Matterport3D**. Black regions denote invalid depth values or masked regions in RGB. Overall, our method shows distinct depth boundaries, especially in the red boxes.

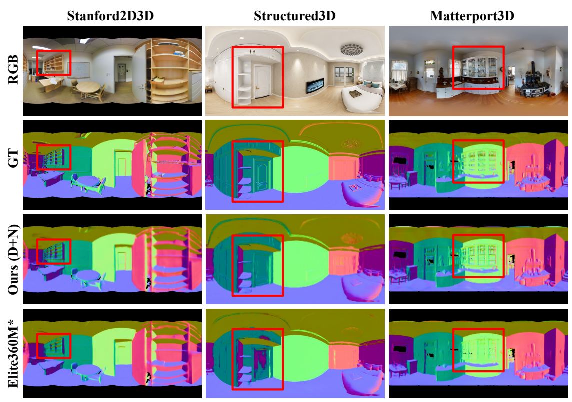

Figure 6. **Qualitative comparison of surface normal estimation**. Our method shows clear and consistent surface normals on the object boundaries and plane region in general.

4.3. Ablation Study

One of our key contributions is that we are the first to integrate SH into holistic panoramic 3D scene understanding. To validate that our SH-based scene understanding framework is an appropriate choice for our target task, we compare depth metrics among different variants of `HUSH` on the Stanford2D3D dataset. Table 2 shows the effects of our design choices, such as queries of fixed SH bases, SH coefficient extractor, SH basis index module, and the degree of SH (*i.e.*, number of SH basis functions). In Fig. 7, we visualize task-wise index maps obtained from our SH basis index module, showing the top-5 dominant queries.

SH for 3D scene understanding. We first validate the usage of SH as queries in our hierarchical attention module by comparing it with learnable queries as in [20, 24], *Method 1* and *Method 2* in Table 2. For a fair comparison, we set the same number of learnable queries. Supporting our observation in Sec. 3.1, `HUSH` framework trained with SH bases performs better than the learnable query-based version. Additionally, we note that in Fig. 7, `HUSH` uses a confined set of SH bases for each task, producing geometrically interpretable results. In contrast, the comparison method with learnable queries requires a larger number of queries to represent the 3D information of a scene and demonstrates less coherence between the queries and geometric structures than ours. More analysis of SH in our network is available in the supplementary material.

SH basis index module. We show the effectiveness of our SH basis index module by comparing *Method 3* and *Full*, where *Full* has improvements on both Abs Rel and RMSE metrics. Interestingly, in Fig. 7, the index map visualization results show coherence on which basis is dominant, depending on the task and the region. For example, in surface normal estimation, the dominant basis highlighted on the floor region and the dominant basis highlighted on the wall region have the same index, respectively, across every example. Similar tendencies are shown in depth estimation as well, where for all examples, dominant bases are clustered corresponding to the depth range. Last but not least, indices highlighted in surface normal estimation and depth estimation are distinctive. This aligns perfectly with our intuition explained in Sec. 3.3, where each basis operates as a structural prior, and different bases are activated depending on the task and the region.

SH basis configuration. Our scene-adaptive SH basis modification is also shown to be effective, where *Full* shows better results than *Method 2* in Table 2. We also ablate on how the degree of SH affects the 3D understanding performance. Our *Full* model utilizes the level of 10 (*i.e.*, 55 SH bases), while *Method 3* in Table 2 uses 5 levels (*i.e.*, 15 SH

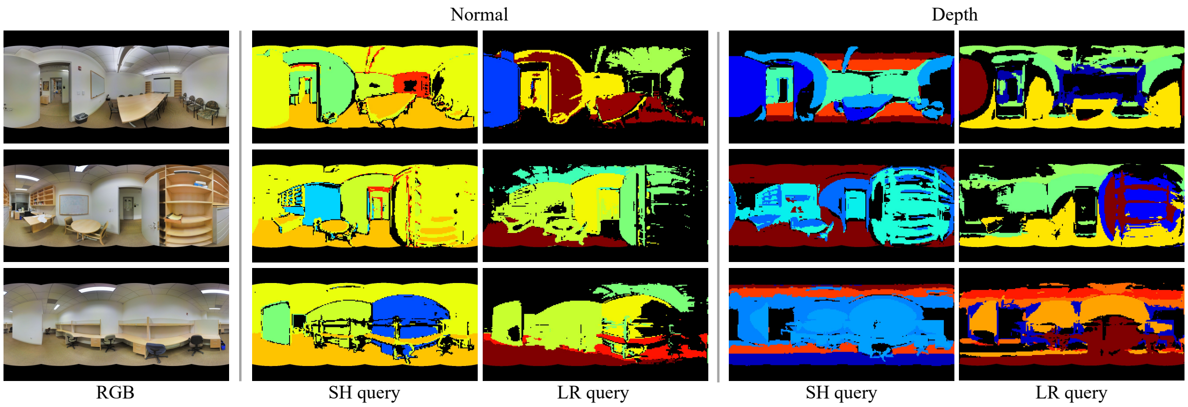

Figure 7. **Task-relevant index map visualization**. We select the top-5 dominant queries from the SH index map for each task. SH query and learnable (LR) query indicate our HUSH framework trained with SH bases and learnable parameters as a query for hierarchical attention module, respectively. By virtue of the scene-adaptive SH bases, our method mainly focuses on the same SH bases on the plane regions where pixels have the same values.

Method	SH	K	IDX	L	Abs Rel↓	Sq Rel↓	RMSE↓	δ_1↑
1			✓	10	0.080	0.056	0.342	0.931
2	✓		✓	10	**0.078**	0.053	0.340	0.937
3	✓	✓		10	0.081	**0.052**	0.336	0.938
4	✓	✓	✓	5	**0.078**	0.056	0.337	**0.940**
Full	✓	✓	✓	10	**0.078**	0.055	**0.333**	0.938

Table 2. **Ablation study on the Stanford2D3D dataset**. SH: SH basis query, K: SH coefficient extractor, IDX: SH basis index module, L: Level of SH basis function.

bases). It is shown that the performance gap between two variants is marginal. It means that our method is comprehensive and robust enough so that it shows similar performance with less number of SH bases. It also addresses that our basis index module successfully guides each SH basis to be more concentrated on separate tasks, thereby reducing ambiguity when using a large number of bases. Finally, it opens room for additional 3D understanding tasks since we may expect that another distinctive set of SH bases will be utilized for the additional task.

4.4. Application: Layout Estimation

We show the extendability of our framework by simply adding a room layout estimation head to HUSH. To successfully train the layout head, we fine-tune the HUSH framework with the layout head using a pre-trained HUSH (D+N) model. Here, we use our HUSH framework trained 100 epochs on the Matterport3D dataset and then fine-tune 200 epochs on the MatterportLayout dataset [42]. As shown in Fig. 8, thanks to our SH-based comprehensive scene understanding framework, the HUSH-based layout estimation network shows comparable results to recent layout estimation-only methods [13, 29, 33] that are trained ∼10× epochs more compared to ours. More qualitative results and training settings for the layout estimation are available in the supplementary material.

Figure 8. **HUSH framework-based layout estimation on the MatterportLayout dataset**. Blue and green lines represent ground truth labels and our predictions, respectively.

5. Conclusion

Based on the observation that panoramic information can be treated as spherical surface function, and can be represented using spherical harmonics, we have proposed a novel Holistic panoramic 3D scene Understanding framework using Spherical Harmonics, dubbed as HUSH. It consists of the SH-based hierarchical attention module and SH basis index module that successfully aggregate the spherical geometry and context information from the SH basis and multi-scale image features. The proposed modules generate comprehensive scene features and task-relevant features, which help to achieve state-of-the-art performance on panorama depth and surface normal estimation and extendability to other 3D scene understanding tasks, such as layout estimation or plane segmentation, via extensive experiments. We suggest a novel way of representing and understanding equirectangular information using spherical harmonics.

Acknowledgments. This work was supported by Institute of Information & communications Technology Planning & Evaluation (IITP) grant funded by the Korea government (MSIT) (No.RS-2022-II220907, Development of AI Bots Collaboration Platform and Self-organizing AI, No.RS-2020-II201336, Artificial Intelligence Graduate School Program (UNIST) and No.RS-2022-II220612, Geometric and Physical Commonsense Reasoning based Behavior Intelligence for Embodied AI).

References

[1] Hao Ai and Lin Wang. Elite360D: Towards Efficient 360 Depth Estimation via Semantic-and Distance-Aware Bi-Projection Fusion. In *CVPR*, 2024. 2, 5, 6

[2] Hao Ai and Lin Wang. Elite360M: Efficient 360 Multi-task Learning via Bi-projection Fusion and Cross-task Collaboration. *arXiv preprint arXiv:2408.09336*, 2024. 2, 3, 6

[3] Hao Ai, Zidong Cao, Yan-Pei Cao, Ying Shan, and Lin Wang. HRDFuse: Monocular 360deg Depth Estimation by Collaboratively Learning Holistic-With-Regional Depth Distributions. In *CVPR*, 2023. 1, 2, 4, 5, 6

[4] Iro Armeni, Sasha Sax, Amir R Zamir, and Silvio Savarese. Joint 2d-3d-semantic data for indoor scene understanding. *arXiv preprint arXiv:1702.01105*, 2017. 5, 6

[5] Shariq Farooq Bhat, Ibraheem Alhashim, and Peter Wonka. AdaBins: Depth estimation using adaptive bins. In *CVPR*, 2021. 2

[6] Shariq Farooq Bhat, Ibraheem Alhashim, and Peter Wonka. LocalBins: Improving depth estimation by learning local distributions. In *ECCV*, 2022. 2

[7] Angel Chang, Angela Dai, Thomas Funkhouser, Maciej Halber, Matthias Niessner, Manolis Savva, Shuran Song, Andy Zeng, and Yinda Zhang. Matterport3D: Learning from RGB-D data in indoor environments. *arXiv preprint arXiv:1709.06158*, 2017. 5, 6

[8] James M Coughlan and Alan L Yuille. Manhattan world: Compass direction from a single image by bayesian inference. In *ICCV*, 1999. 3

[9] Yuan Dong, Chuan Fang, Liefeng Bo, Zilong Dong, and Ping Tan. PanoContext-Former: Panoramic total scene understanding with a transformer. In *CVPR*, 2024. 2, 3

[10] Marc Eder, Pierre Moulon, and Li Guan. Pano popups: Indoor 3d reconstruction with a plane-aware network. In *3DV*, 2019. 2

[11] Darya Frolova, Denis Simakov, and Ronen Basri. Accuracy of spherical harmonic approximations for images of lambertian objects under far and near lighting. In *ECCV*, 2004. 3

[12] Hualie Jiang, Zhe Sheng, Siyu Zhu, Zilong Dong, and Rui Huang. Unifuse: Unidirectional fusion for 360 panorama depth estimation. *RA-L*, 6(2):1519–1526, 2021. 1, 2, 6

[13] Zhigang Jiang, Zhongzheng Xiang, Jinhua Xu, and Ming Zhao. Lgt-net: Indoor panoramic room layout estimation with geometry-aware transformer network. In *CVPR*, 2022. 8

[14] Kyungdon Joo, Tae-Hyun Oh, Junsik Kim, and In So Kweon. Robust and globally optimal manhattan frame estimation in near real time. *IEEE TPAMI*, 2018. 3

[15] Kyungdon Joo, Tae-Hyun Oh, In So Kweon, and Jean-Charles Bazin. Globally optimal inlier set maximization for atlanta world understanding. *IEEE TPAMI*, 2019. 3

[16] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3d gaussian splatting for real-time radiance field rendering. *ACM TOG*, 2023. 3

[17] Iro Laina, Christian Rupprecht, Vasileios Belagiannis, Federico Tombari, and Nassir Navab. Deeper depth prediction with fully convolutional residual networks. In *3DV*, 2016. 5

[18] Jongsung Lee, Gyeongsu Cho, Jeongin Park, Kyongjun Kim, Seongoh Lee, Jung-Hee Kim, Seong-Gyun Jeong, and Kyungdon Joo. SlaBins: Fisheye depth estimation using slanted bins on road environments. In *ICCV*, 2023. 2

[19] Yuyan Li, Yuliang Guo, Zhixin Yan, Xinyu Huang, Ye Duan, and Liu Ren. OmniFusion: 360 monocular depth estimation via geometry-aware fusion. In *CVPR*, 2022. 1, 2, 5

[20] Li Liu, Ruijie Zhu, Jiacheng Deng, Ziyang Song, Wenfei Yang, and Tianzhu Zhang. Plane2depth: Hierarchical adaptive plane guidance for monocular depth estimation. *IEEE TCSVT*, 2024. 2, 4, 7

[21] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *ICCV*, 2021. 3

[22] Ilya Loshchilov and Frank Hutter. SGDR: Stochastic Gradient Descent with Warm Restarts. *arXiv preprint arXiv:1608.03983*, 2017. 5

[23] Ilya Loshchilov and Frank Hutter. Decoupled Weight Decay Regularization. In *ICLR*, 2018. 5

[24] Luigi Piccinelli, Christos Sakaridis, and Fisher Yu. iDisc: Internal discretization for monocular depth estimation. In *CVPR*, 2023. 2, 7

[25] Ravi Ramamoorthi and Pat Hanrahan. An efficient representation for irradiance environment maps. In *Proceedings of the 28th annual conference on Computer graphics and interactive techniques*, 2001. 2, 3

[26] Manuel Rey-Area, Mingze Yuan, and Christian Richardt. 360monodepth: High-resolution 360deg monocular depth estimation. In *CVPR*, 2022. 1, 2

[27] Grant Schindler and Frank Dellaert. Atlanta world: An expectation maximization framework for simultaneous low-level edge grouping and camera calibration in complex man-made environments. In *CVPR*, 2004. 3

[28] Zhijie Shen, Chunyu Lin, Kang Liao, Lang Nie, Zishuo Zheng, and Yao Zhao. PanoFormer: Panorama Transformer for Indoor 360 Depth Estimation. In *ECCV*, 2022. 1, 2, 6

[29] Zhijie Shen, Zishuo Zheng, Chunyu Lin, Lang Nie, Kang Liao, Shuai Zheng, and Yao Zhao. Disentangling orthogonal planes for indoor panoramic room layout estimation with cross-scale distortion awareness. In *CVPR*, 2023. 8

[30] Zhijie Shen, Chunyu Lin, Lang Nie, and Kang Liao. Revisiting 360 depth estimation with panogabor: A new fusion perspective. *arXiv preprint arXiv:2408.16227*, 2024. 5

[31] Leslie N Smith and Nicholay Topin. Super-convergence: Very fast training of neural networks using large learning rates. In *Artificial intelligence and machine learning for multi-domain operations applications*, pages 369–386. SPIE, 2019. 5

[32] Cheng Sun, Min Sun, and Hwann-Tzong Chen. HoHoNet: 360 indoor holistic understanding with latent horizontal features. In *CVPR*, 2021. 2

[33] Yu-Ju Tsai, Jin-Cheng Jhang, Jingjing Zheng, Wei Wang, Albert YC Chen, Min Sun, Cheng-Hao Kuo, and Ming-Hsuan Yang. No more ambiguity in 360deg room layout via bi-layout estimation. In *CVPR*, 2024. 8

[34] Fu-En Wang, Yu-Hsuan Yeh, Min Sun, Wei-Chen Chiu, and Yi-Hsuan Tsai. BiFuse: Monocular 360 depth estimation via bi-projection fusion. In *CVPR*, 2020. 1, 2

[35] Suttisak Wizadwongsa, Pakkapon Phongthawee, Jiraphon Yenphraphai, and Supasorn Suwajanakorn. Nex: Real-time view synthesis with neural basis expansion. In *CVPR*, 2021. 3

[36] Chenglei Wu, Bennett Wilburn, Yasuyuki Matsushita, and Christian Theobalt. High-quality shape from multi-view stereo and shading under general illumination. In *CVPR*, 2011. 3

[37] Alex Yu, Ruilong Li, Matthew Tancik, Hao Li, Ren Ng, and Angjoo Kanazawa. Plenoctrees for real-time rendering of neural radiance fields. In *ICCV*, 2021. 3

[38] Haozheng Yu, Lu He, Bing Jian, Weiwei Feng, and Shan Liu. PanelNet: Understanding 360 Indoor Environment via Panel Representation. In *CVPR*, 2023. 1, 2, 3

[39] Ilwi Yun, Chanyong Shin, Hyunku Lee, Hyuk-Jae Lee, and Chae Eun Rhee. EGformer: Equirectangular Geometry-biased Transformer for 360 Depth Estimation. In *ICCV*, 2023. 1, 2, 6

[40] Jia Zheng, Junfei Zhang, Jing Li, Rui Tang, Shenghua Gao, and Zihan Zhou. Structured3d: A large photo-realistic dataset for structured 3d modeling. In *ECCV*, 2020. 5, 6

[41] Xizhou Zhu, Weijie Su, Lewei Lu, Bin Li, Xiaogang Wang, and Jifeng Dai. Deformable detr: Deformable transformers for end-to-end object detection. *ICLR*, 2021. 3

[42] Chuhang Zou, Jheng-Wei Su, Chi-Han Peng, Alex Colburn, Qi Shan, Peter Wonka, Hung-Kuo Chu, and Derek Hoiem. Manhattan room layout reconstruction from a single 360 image: A comparative study of state-of-the-art methods. *IJCV*, 2021. 8